News-Based Exchange-Rate Risk, Market Alignment, and Volatility Dynamics across African Economies

Continental Panels, GARCH-Family Heterogeneity, Dynamic Transmission, and Out-of-Sample Forecast Evidence

Abstract

We introduce a market-alignment framework for organizing cross-country heterogeneity in news-based exchange-rate risk transmission and separate three empirical objects that are often conflated: information content, conditional-variance transmission, and forecast value. Using a 51-country African panel drawn from 54 source markets over January 2015–December 2025, we show that the News-Based Exchange-Rate Risk Index (NBERI) carries leading information about subsequent FX volatility, that transmission is stronger in markets where the FX-risk narrative is closely aligned with the bilateral exchange-rate level, and that the conditional-variance channel is selective rather than universal. In the continental benchmark, lagged NBERI predicts next-month volatility after controlling for volatility history, returns, inflation, country effects, and common time shocks ($\beta = 0.030$, $p = 0.004$), and remains significant after shared currencies are collapsed to 36 market clusters. Full-panel interaction tests imply an aligned-group slope of 0.063 ($p = 0.007$) versus 0.017 ($p = 0.147$) elsewhere, while a 1,000-draw random-subset benchmark places the selected group's effect near the upper tail of comparably sized groups. In the 15 aligned markets, the horizon-1 local-projection response is 0.121 ($p < 0.001$), NBERI explains 3.39 percent of 12-month volatility forecast-error variance, and adding NBERI improves 2023–2025 pooled RMSE by 0.89 percent. Profile-likelihood GARCH-family tests with false-discovery-rate control identify robust conditional-variance effects only in Ghana and Libya. The evidence separates broad leading information from selective variance transmission.

Keywords: news-based risk; exchange rates; volatility; market alignment; GARCH-X; local projections; Africa
JEL Classification: F31, C22, C23, C53, G15

1. Introduction

Exchange-rate instability is a recurrent source of macroeconomic risk across African economies. Depreciation passes into import prices, changes the domestic-currency value of external liabilities, complicates monetary policy, and can intensify balance-sheet and liquidity pressures. Many conventional surveillance indicators are observed only after these pressures have appeared in exchange rates, reserves, inflation, or external financing conditions. A risk measure that consistently moves before observed currency stress can be useful even if it does not replace market-based volatility models.

News-based indicators provide one route to earlier measurement. Economic reporting aggregates dispersed information about foreign-exchange availability, reserve adequacy, external financing, policy adjustments, debt negotiations, devaluation expectations, and market access. The broader uncertainty literature shows that disciplined text-based indices can contain macroeconomic information that is not reducible to contemporaneous market movements (Baker et al., 2016; Caldara and Iacoviello, 2022). In foreign-exchange markets, the economic content of news is especially plausible because prices respond rapidly to changes in the information set (Andersen et al., 2003). The empirical question is whether a focused exchange-rate risk index adds information after the currency market's own history and macroeconomic controls are taken into account.

This paper studies the News-Based Exchange-Rate Risk Index (NBERI), a monthly measure of foreign-exchange and external-sector risk narratives. We treat the index strictly as an information-state variable: it is not market volatility, not an exchange-rate transformation, and not a mechanical restatement of realized depreciation.[1] The empirical question is whether the information environment summarized by NBERI precedes subsequent

[1]The distinction is not semantic: NBERI is constructed from news narratives, while every market outcome in the paper is constructed from observed bilateral exchange-rate data.

exchange-rate instability after the market's own history and conventional macroeconomic controls are taken into account.

We organize cross-country heterogeneity with a market-alignment screen. The screen ranks eligible markets by the contemporaneous correlation between the FX-risk narrative and the bilateral exchange-rate level, and we test in the full panel whether the aligned group shows stronger risk-to-volatility transmission. The selected 15-country group is also benchmarked against 1,000 random 15-country subsets. The ranking is descriptive, not causal. The interaction tests, not the ranking itself, carry the formal evidence.[2]

We keep three objects apart: information content, conditional-variance transmission, and forecast value. Two-way fixed effects, PVARs, and local projections ask whether NBERI carries leading information about future instability. GARCH-X, EGARCH-X, and GJR/TGARCH-X ask whether NBERI enters the conditional-variance recursion once endogenous volatility persistence is absorbed. Expanding-window forecasts ask whether the signal reduces genuine out-of-sample loss. These need not agree. When they do not, the divergence itself is informative about how news-based FX risk enters currency markets.[3]

The panel has 54 countries, of which 51 pass a coverage-and-variation screen. A 36-market currency-cluster robustness design prevents repeated CFA and Common Monetary Area paths from being counted as independent evidence.[4] The broad panel establishes the common leading-information result before the analysis turns to heterogeneity.

The continental coefficient is 0.030 ($p = 0.004$). The aligned-group full-panel slope is 0.063 ($p = 0.007$); the remaining-market slope is 0.017 ($p = 0.147$). At horizon 1, the local-projection increment for the aligned group is 0.072 ($p = 0.019$), and the selected group's fixed-effects effect sits near the 94th percentile of 1,000 random 15-country subsets. Conditional-variance evidence is narrower. After profile-likelihood refinement and within-family false-discovery control, only Ghana under EGARCH-X and Libya under GARCH-X show robust NBERI variance effects. Out-of-sample gains are positive but small. The paper separates broad news-based information, alignment-dependent transmission, and market-specific variance effects.

2. Related Literature and Testable Hypotheses

2.1 News, information, and exchange-rate risk

Newspaper-based uncertainty measures sit between realized market variables and survey expectations. Baker et al. (2016) show that a transparent coding rule can produce an uncertainty index with macroeconomic content; Caldara and Iacoviello (2022) show that a focused index can isolate a single class of risk. Neither result implies that media coverage causes asset-price volatility. The point is informational: news can aggregate policy signals, funding conditions, geopolitical developments, and expectations before those forces show up in observed outcomes.

Foreign-exchange markets provide a demanding setting for such a test. Andersen et al. (2003) document rapid exchange-rate responses to macroeconomic information, emphasizing that the surprise component of news matters for price discovery. Across African economies, information transmission is further shaped by heterogeneous exchange-rate regimes, market depth, intervention practices, foreign-exchange availability, and the presence of currency unions. A useful empirical design must therefore allow both a common continental signal and cross-market heterogeneity. Assigning a separate narrative to every country coefficient is not useful.

2.2 Conditional volatility and panel dependence

ARCH and GARCH models remain the standard parametric framework for persistent conditional heteroskedasticity (Engle, 1982; Bollerslev, 1986). EGARCH permits asymmetric responses in log variance (Nelson, 1991), while

[2] Market alignment is not used as an instrument, treatment, or structural parameter. The full-panel interactions and the random-subset benchmark are the formal tests of whether the aligned group shows unusually strong transmission.

[3] A variable can forecast future realized volatility without entering a particular parametric conditional-variance recursion, and an in-sample predictive relation can exist without producing economically large out-of-sample forecast gains.

[4] This adjustment matters for the WAEMU/CEMAC CFA zones and the tightly linked Common Monetary Area currencies, where repeated country labels would otherwise overstate the effective cross-section.

GJR-GARCH allows negative and positive innovations to affect conditional variance differently (Glosten et al., 1993). GARCH-X extensions provide a direct test of whether an observed state variable contributes to conditional variance after endogenous volatility persistence is absorbed (Francq and Thieu, 2019; Pedersen and Rahbek, 2019). In our setting, the distinction matters: a news-risk measure can predict next-month instability without entering every currency's variance recursion through the same parameter.

Panel settings add a second complication: contemporaneous shocks and common institutions generate cross-sectional dependence. Cermeño and Grier (2006) demonstrate the usefulness of panel approaches to conditional heteroskedasticity, while Driscoll and Kraay (1998) provide covariance estimation that remains robust to broad forms of cross-sectional and temporal dependence. These considerations are especially relevant in Africa because the CFA franc zones and the Common Monetary Area create mechanically related exchange-rate paths. We therefore combine country effects and common time effects with Driscoll-Kraay inference and report a currency-cluster robustness design rather than treating all country observations as independent market realizations.

2.3 Dynamic propagation and forecast evaluation

A statistically significant one-step coefficient does not establish the duration of a predictive relationship. Panel VAR methods summarize feedback within a multivariate system, while generalized impulse responses avoid dependence on an arbitrary Cholesky ordering (Pesaran and Shin, 1998). Local projections provide a complementary horizon-by-horizon representation that is less dependent on the recursive propagation of a single estimated transition matrix (Jordà, 2005). We use the two approaches jointly because they answer related but distinct questions.

Forecast evaluation imposes a stronger requirement. A variable can be significant in sample yet fail to reduce out-of-sample loss. Clark and West (2007) develop tests for nested predictive models, and Patton (2011) shows why loss functions such as QLIKE are useful when the volatility target is measured with noise. These insights motivate an explicit horse race between a benchmark built from exchange-rate history and inflation and an augmented model that adds lagged NBERI.

2.4 Hypotheses

H1 - Leading-risk hypothesis. Conditional on lagged volatility, lagged exchange-rate returns, inflation, country effects, and common time shocks, a higher lagged NBERI state should predict a higher subsequent monthly FX volatility proxy.

H2 - Dynamic-transmission hypothesis. A positive NBERI innovation should be followed by a positive short-horizon response of the FX volatility proxy, and the dynamic contribution should attenuate over time.

H3 - Incremental-forecast hypothesis. Adding lagged NBERI to a nested benchmark containing exchange-rate history and inflation should reduce out-of-sample forecast loss.

H4 - Market-alignment heterogeneity hypothesis. The leading NBERI-volatility relationship should be stronger in markets where the FX-risk pillar is more closely aligned with the observed exchange-rate state, but the alignment screen need not imply a monotonic or universal GARCH-X effect.

3. Data and Empirical Design

3.1 Continental sample and objective coverage screen

The empirical dataset combines the monthly NBERI panel with harmonized monthly bilateral exchange-rate and inflation panels supplied for the study. The common estimation window is January 2015 through December 2025. We end the sample in December 2025 to keep a consistent macroeconomic control set and a fully defined out-of-sample evaluation period, even though some source data extend further.

The source exchange-rate file contains 54 African countries. Inclusion is determined before model estimation using three transparent requirements: at least 60 usable adjacent monthly exchange-rate returns; at least 5 percent of usable months with nonzero exchange-rate changes; and nonzero return variance. Fifty-one countries satisfy these conditions. Djibouti and Eritrea are excluded because the supplied official-rate series are effectively invariant

over the sample, while Somalia has only 41 usable adjacent monthly returns. A documented unit discontinuity in Zimbabwe in April 2024 is treated as missing, not as an economic return. We do not interpolate missing returns.

Table 1. Continental sample construction

Item	Count	Definition
Source countries	54	All African country series in supplied monthly FX panel
Eligible countries	51	Pass coverage and return-variation screen
North Africa	6	Eligible countries
West Africa	16	Eligible countries
Central Africa	9	Eligible countries
East Africa	11	Eligible countries
Southern Africa	9	Eligible countries
Currency-market clusters	36	Robustness sample after shared/linked currency collapse

Notes: The currency-cluster robustness collapses WAEMU-XOF, CEMAC-XAF, and CMA-ZAR-linked observations so that repeated or tightly linked currency paths do not mechanically increase the effective cross-section.

3.2 Variables and measurement

The bilateral exchange rate S_{it} is expressed as units of local currency per U.S. dollar. A positive return therefore represents a local-currency depreciation. Monthly log returns are defined as:

$$r_{it} = 100 \left[\ln(S_{it}) - \ln(S_{i,t-1}) \right]. \tag{1}$$

The monthly volatility outcome used in the panel, PVAR, local-projection, and forecast exercises is $v_{it} = \ln(1 + r_{it}^2)$. This transformation reduces the influence of extreme squared returns while retaining the ordering of large currency movements. One monthly exchange-rate observation cannot recover intramonth quadratic variation, so we call v_it a monthly volatility proxy rather than realized volatility.[5]

NBERI is the monthly news-based exchange-rate risk score derived from the exchange-rate/external-risk component of the study's news classification system. It measures the intensity of reported foreign-exchange and external-sector risk narratives and is kept separate from the exchange-rate outcome. To prevent persistent cross-country differences in scale from driving pooled coefficients, NBERI, the volatility proxy, returns where indicated, and inflation are standardized within country. The principal NBERI coefficient is therefore interpretable in standard-deviation units. Inflation enters as a conventional macroeconomic control, not as part of NBERI.

Table 2. Descriptive statistics, 2015-2025

Variable	N	Mean	SD	P25	Median	P75	Max
NBERI	6,731	5.443	14.410	0.271	1.640	4.967	211.129
FX return (%)	6,446	0.707	5.109	-0.468	0.183	1.105	133.715
$\ln(1 + r^2)$	6,446	0.973	1.226	0.070	0.492	1.477	9.791
Inflation (%)	6,632	11.252	37.600	2.158	4.900	9.662	820.770

Notes: FX return is the monthly log change in local currency per U.S. dollar. The volatility proxy is $\ln(1 + r^2)$. Raw descriptive statistics are shown before within-country standardization.

[5]The terminology is conservative: monthly squared returns preserve the ordering of large currency movements but cannot recover high-frequency realized variance.

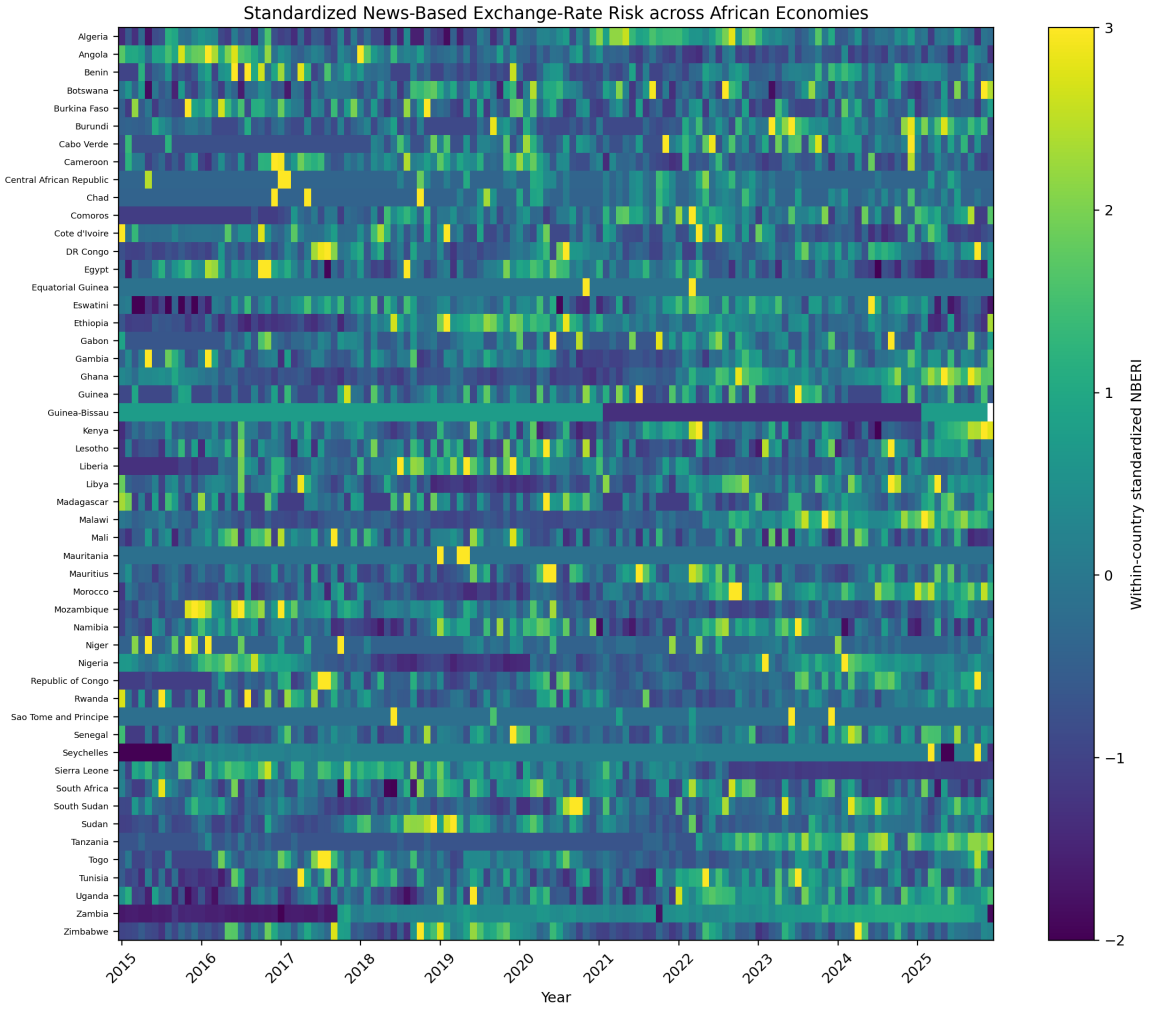


Figure 1: Standardized monthly NBERI across eligible African economies, 2015–2025.

3.3 Shared currencies and effective market dependence

A country panel is not a panel of independent currencies. The WAEMU members in the sample share the West African CFA franc; the CEMAC members share the Central African CFA franc; and South Africa, Lesotho, Eswatini, and Namibia are treated as a linked Common Monetary Area cluster for robustness. The principal regressions retain country observations because NBERI is country-specific and the research question concerns country risk environments. A complementary 36-cluster specification collapses shared or linked exchange-rate paths so that the same currency movement is not counted repeatedly as independent market evidence. Cross-sectional dependence in the full panel is addressed further through common time effects and Driscoll-Kraay covariance estimation.

3.4 Two-way fixed-effects benchmark

The principal specification asks whether the current risk state contains information about next-month exchange-rate instability after conditioning on market history and inflation:

$$v_{it} = \alpha_i + \lambda_t + \beta\, NBERI_{i,t-1} + \rho v_{i,t-1} + \theta r_{i,t-1} + \kappa \pi_{i,t-1} + \varepsilon_{it}. \tag{2}$$

Country effects absorb time-invariant differences in market structure, reporting intensity, currency regime, and level effects. Calendar-time effects absorb shocks shared across the continent. Inference uses Driscoll-Kraay standard errors with a four-month bandwidth, allowing heteroskedasticity, serial correlation, and broad cross-sectional dependence. We re-estimate the same model on the 36 currency-market clusters. A threshold extension interacts lagged NBERI with an indicator for observations above each country's 75th percentile to test whether high-risk states amplify the slope.

3.5 Pooled panel GARCH-X robustness

The conditional-volatility exercise is secondary. The exchange-rate series are monthly, but the exercise still provides a useful robustness test of whether NBERI shifts a common GARCH variance recursion after volatility persistence is modeled directly. Monthly returns are standardized within country, and the pooled likelihood allows a common GARCH persistence structure across countries with Student-t innovations. The nested GARCH(1,1) benchmark is

$$h_{it}^{\text{GARCH}} = \omega + \alpha \varepsilon_{i,t-1}^2 + \beta h_{i,t-1}. \tag{3}$$

The GARCH-X specification augments that recursion multiplicatively with lagged NBERI:

$$h_{it}^{\text{GARCH-X}} = h_{it}^{\text{GARCH}} \exp\left(\delta\, NBERI_{i,t-1}\right). \tag{4}$$

The coefficient δ is the parameter of interest. A positive significant delta would indicate that the news-risk state shifts the common conditional-variance recursion beyond lagged squared innovations and lagged variance. We compare the baseline GARCH, GARCH-X, GARCH-X with inflation, and regional-slope GARCH-X models by likelihood and information criteria. A pooled GJR-GARCH-X specification is reported in the appendix to test whether a common threshold response to negative innovations is required.

3.6 Fixed-effects panel VAR and generalized dynamics

Dynamic feedback is summarized with a parsimonious fixed-effects PVAR(1) containing standardized NBERI, FX return, and the FX volatility proxy, with lagged inflation as an exogenous control. Each equation includes country and calendar-time effects:

$$Y_{it} = AY_{i,t-1} + B\pi_{i,t-1} + \mu_i + \tau_t + u_{it}, \qquad Y_{it} = \begin{bmatrix} NBERI_{it} & r_{it} & v_{it} \end{bmatrix}'. \tag{5}$$

The long time dimension and 51-country cross-section make an aggressively instrumented GMM system unnecessary for our predictive purpose and would create a material risk of instrument proliferation. Coefficient inference therefore uses Driscoll-Kraay covariance estimates. Generalized impulse responses and generalized

forecast-error variance decompositions are computed from the reduced-form innovation covariance matrix so that the reported dynamics are invariant to variable ordering (Pesaran and Shin, 1998). Confidence bands for generalized responses use 300 country-block bootstrap replications. Stability is evaluated from the companion-matrix eigenvalues.

3.7 Panel local projections

Local projections provide a complementary horizon-specific test (Jordà, 2005). For horizons $h = 0, \ldots, 12$, the estimating equation is

$$v_{i,t+h} = \alpha_i^{(h)} + \lambda_t^{(h)} + \beta_h \, NBERI_{i,t} + \rho_h v_{i,t-1} + \theta_h r_{i,t-1} + \kappa_h \pi_{i,t} + \varepsilon_{i,t+h}^{(h)}. \tag{6}$$

Thus, β_h traces the horizon-specific predictive response of the standardized volatility proxy to the current standardized NBERI state while holding fixed lagged volatility, lagged return, current inflation, country effects, and calendar-time effects. Driscoll-Kraay standard errors are used at each horizon. We interpret the coefficient sequence in Equation (6) as a predictive response profile, not a structural causal impulse response.

3.8 Expanding-window out-of-sample horse race

The forecast experiment covers January 2023 through December 2025. Define the forecast target as $\ell_{it} \equiv \ln(0.01+r_{it}^2)$. At each forecast origin, the benchmark and NBERI-augmented one-step-ahead forecasts can be written compactly as

$$\widehat{\ell}^{\mathrm{BM}}_{i,t+1|t} = f\left(\ell_{it}, \ell_{i,t-1}, r_{it}, \pi_{it}, \pi_{i,t-1}, \alpha_i\right), \tag{7}$$

$$\widehat{\ell}^{\mathrm{AUG}}_{i,t+1|t} = g\left(\ell_{it}, \ell_{i,t-1}, r_{it}, \pi_{it}, \pi_{i,t-1}, NBERI_{it}, NBERI_{i,t-1}, \alpha_i\right). \tag{8}$$

Here, $f(\cdot)$ and $g(\cdot)$ denote the estimated linear forecasting maps with country intercepts; Equation (8) nests the same market-history and inflation information set as Equation (7) and adds two NBERI lags. Country standardization parameters use only training data available at each forecast origin, so no future information enters the 2023–2025 evaluation window.[6] Forecasts are evaluated using RMSE and QLIKE. The Clark-West test assesses nested mean-squared predictive accuracy, and a one-sided HAC test evaluates the monthly QLIKE loss differential (Clark and West, 2007; Patton, 2011).

3.9 Market-alignment extension and country-level GARCH-family design

The continental analysis remains the primary specification. The heterogeneity extension ranks the 51 eligible countries by the market-alignment statistic

$$A_i = \mathrm{Corr}(FXExternalRisk_{it}, S_{it}), \qquad t \in \mathcal{T} = \{2015{:}01, \ldots, 2025{:}12\}, \tag{9}$$

where S_{it} is the bilateral exchange-rate level measured as local-currency units per U.S. dollar. The 15 highest-alignment markets are Zambia, Ghana, Malawi, Tanzania, Algeria, Burundi, Libya, Nigeria, Tunisia, Morocco, Liberia, Namibia, Congo DRC, Guinea, and Benin. A positive value of A_i means elevated FX-risk narratives tend to occur when the local-currency price of the U.S. dollar is high. It says nothing about next-month depreciation.[7]

We use the level-correlation ranking only to organize heterogeneity. To guard against a purely mechanical subsample story, we estimate the interaction NBERI_i,t-1 × Aligned_i in the full eligible panel, compare the selected-group slope with the remaining-market slope, estimate the same interaction in the horizon-1 local projection,

[6] Re-estimating standardization parameters at each forecast origin matters: using full-sample means or standard deviations would leak future information into the evaluation window.

[7] The bilateral exchange rate is measured as units of local currency per U.S. dollar. A higher level therefore denotes a larger domestic-currency price of the dollar, but the level correlation in Equation (9) does not by itself identify the direction or timing of subsequent depreciation.

evaluate a continuous NBERI-by-alignment interaction, and benchmark the selected-group coefficient against 1,000 random 15-country subsets. Spearman and detrended log-exchange-rate correlations are reported as descriptive sensitivity checks. The interaction tests, not the raw ranking itself, carry the formal evidence on differential transmission.[8]

Within the aligned subsample, we rerun the fixed-effects, PVAR, generalized-response, FEVD, local-projection, forecast, absolute-return, and first-difference-NBERI exercises. Country-level conditional variance is then examined using six models: GARCH(1,1), GARCH-X, EGARCH(1,1), EGARCH-X, GJR-GARCH/TGARCH(1,1), and GJR-GARCH/TGARCH-X. Each model uses the AR(1) conditional mean

$$r_{it} = \mu_i + \phi_i r_{i,t-1} + \varepsilon_{it}, \qquad z_{it} = \frac{\varepsilon_{it}}{\sqrt{h_{it}}}, \tag{10}$$

with standardized Student-t innovations. The NBERI-augmented GARCH(1,1) recursion is

$$h_{it} = \left(\omega_i + \alpha_i \varepsilon_{i,t-1}^2 + \beta_i h_{i,t-1}\right) \exp\left(\delta_i NBERI_{i,t-1}\right). \tag{11}$$

For EGARCH-X, the log-variance recursion is

$$\begin{aligned} \ln h_{it} = \omega_i + \beta_i \ln h_{i,t-1} + \alpha_i\left(|z_{i,t-1}| - \mathrm{E}|z_{i,t-1}|\right) \\ + \gamma_i z_{i,t-1} + \delta_i NBERI_{i,t-1}. \end{aligned} \tag{12}$$

For GJR-GARCH/TGARCH-X, the threshold recursion is

$$\begin{aligned} h_{it} = \Big[\omega_i + \left(\alpha_i + \gamma_i \mathbf{1}\{\varepsilon_{i,t-1} < 0\}\right)\varepsilon_{i,t-1}^2 \\ + \beta_i h_{i,t-1}\Big] \exp\left(\delta_i NBERI_{i,t-1}\right). \end{aligned} \tag{13}$$

The corresponding non-X GARCH, EGARCH, and GJR/TGARCH models are obtained by imposing $\delta_i = 0$. Thus, δ_i is the common object of interest across the three families: a positive value means that a higher lagged FX-risk state raises conditional variance after the family's own persistence and innovation terms have been absorbed. Likelihood-ratio tests compare each X model with its nested non-X family, and Benjamini-Hochberg q-values control multiplicity across the 15 countries within each family (Benjamini and Hochberg, 1995).[9]

Nonlinear volatility likelihoods can be sensitive to starting values. We check each reported country-level X estimate with an explicit profile-likelihood search over the NBERI variance coefficient before joint refinement. Let $\vartheta_{-\delta}$ denote all parameters other than δ. The profile estimator is

$$\widehat{\delta}_{\text{prof}} = \arg\max_{\delta} \; \ell\left(\widehat{\vartheta}_{-\delta}(\delta), \delta\right), \qquad \widehat{\vartheta}_{-\delta}(\delta) = \arg\max_{\vartheta_{-\delta}} \ell(\vartheta_{-\delta}, \delta). \tag{14}$$

The profile step in Equation (14) ensures that a reported X effect improves the corresponding nested family, not just a single local optimum. Residual squared-autocorrelation and ARCH-LM diagnostics are reported alongside the likelihood tests.[10]

[8] The ranking organizes the heterogeneity; the interaction estimates test whether the subsequent risk-to-volatility relation actually differs. This keeps the contribution from resting mechanically on a same-sample level correlation.

[9] Writing the three X recursions explicitly makes clear that the NBERI term is tested against different volatility dynamics: symmetric persistence in GARCH-X, logarithmic and sign-sensitive dynamics in EGARCH-X, and threshold asymmetry in GJR/TGARCH-X.

[10] The country-level GARCH-family estimates use custom maximum likelihood with multiple starts, nested likelihood checks, and the explicit profile search in Equation (14). Full country-family results and residual diagnostics are reported in Appendix H.

4. Empirical Results

4.1 Continental risk variation and exchange-rate behavior

The broader panel reveals pronounced heterogeneity in both news risk and exchange-rate adjustment. NBERI is right-skewed, with a mean of 5.44, a median of 1.64, and a maximum above 200 in the raw scale. Monthly FX returns are likewise heavy-tailed, including infrequent but large devaluations and re-pricing episodes. These features are why we standardize within country in pooled models and use Student-t innovations in the conditional-volatility exercise. The heatmap in Figure 1 also shows that elevated risk episodes are neither synchronized across all countries nor confined to one region, supporting a design that controls for common monthly shocks rather than attributing every co-movement to NBERI.

4.2 Baseline predictive relationship and shared-currency robustness

Table 3. Dynamic panel estimates of subsequent FX volatility

Specification	Regressor	Coefficient	DK SE	p-value	N	Within R2
Baseline (no inflation)	Lag NBERI	0.034	0.011	0.001	6,393	0.104
Baseline (no inflation)	Lag volatility	0.288	0.034	<0.001	6,393	0.104
Baseline (no inflation)	Lag return	0.067	0.032	0.034	6,393	0.104
Baseline + inflation	Lag NBERI	0.030	0.010	0.004	6,347	0.107
Baseline + inflation	Lag volatility	0.273	0.035	<0.001	6,347	0.107
Baseline + inflation	Lag return	0.075	0.032	0.021	6,347	0.107
Baseline + inflation	Lag inflation	0.073	0.019	<0.001	6,347	0.107
Currency-cluster robustness	Lag NBERI	0.033	0.013	0.011	4,406	0.154
Currency-cluster robustness	Lag volatility	0.333	0.026	<0.001	4,406	0.154
Currency-cluster robustness	Lag return	0.069	0.029	0.018	4,406	0.154
Currency-cluster robustness	Lag inflation	0.087	0.021	<0.001	4,406	0.154

Notes: Dependent variable is the within-country standardized monthly volatility proxy. All specifications include country and calendar-time effects. The currency-cluster model uses 36 market clusters rather than 51 country series. Driscoll-Kraay bandwidth = 4.

The main result is positive and precise. In the specification that includes inflation, lagged NBERI has a coefficient of 0.030 (Driscoll-Kraay SE = 0.010; p = 0.004). A one-standard-deviation increase in the country-specific risk state is followed by approximately 0.03 standard deviations more exchange-rate volatility in the next month, conditional on prior volatility, prior returns, inflation, country effects, and common time shocks. The effect is smaller than the autoregressive volatility coefficient (0.273), which is consistent with NBERI adding information rather than dominating market history.

The shared-currency robustness matters for interpretation. Once common or tightly linked currency paths are collapsed to 36 market clusters, the NBERI coefficient is 0.033 (SE = 0.013; p = 0.011). The estimated relationship does not depend on treating the same CFA or Common Monetary Area exchange-rate path as multiple independent market realizations. Lagged inflation also remains positive, and lagged returns retain incremental information.

4.3 Regional heterogeneity and nonlinear risk states

Table 4. Regional NBERI slopes and high-risk threshold test

Region / threshold term	Coefficient	DK SE	p-value
North	0.090	0.032	0.005
West	0.021	0.020	0.292
Central	-0.004	0.020	0.839
East	0.014	0.023	0.543
Southern	0.065	0.032	0.042
Wald: equality of five regional slopes	6.217	df = 4	0.184
Lag NBERI below country q75	0.050	0.024	0.036
Increment above country q75	-0.030	0.031	0.339
Total high-risk slope	0.020	0.014	0.149

Notes: Regional slopes are estimated jointly with the same dynamic controls, country effects, and time effects as the baseline model. The Wald row tests equality of all five regional NBERI slopes. The threshold is each country's 75th percentile of NBERI.

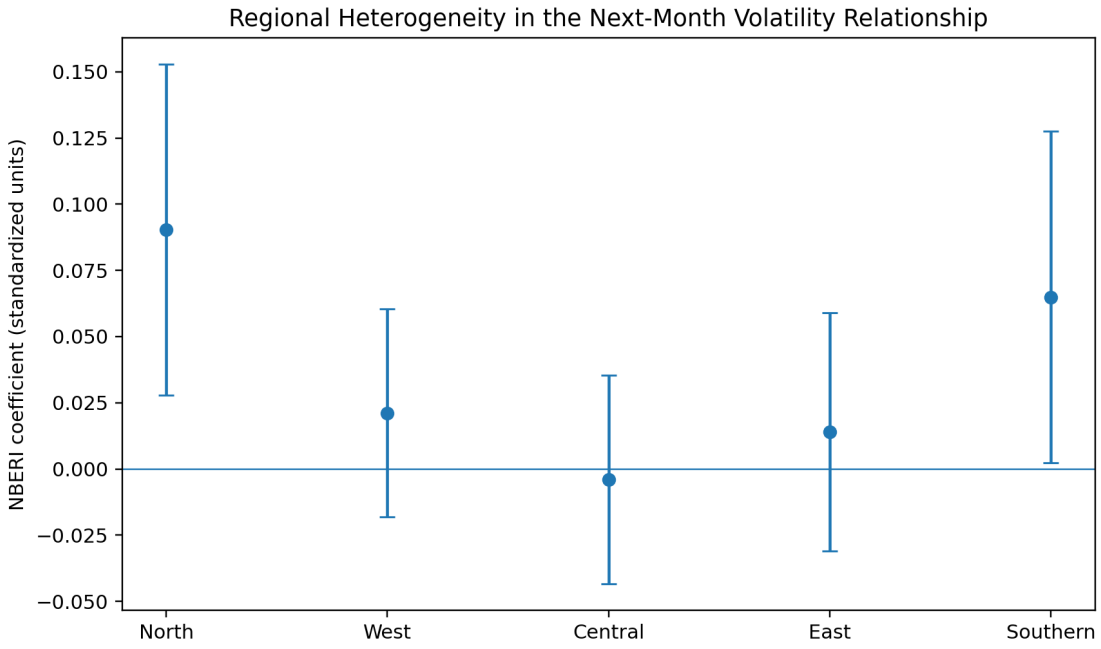


Figure 2: Regional heterogeneity in the next-month NBERI–volatility relationship.

North Africa (0.090, p = 0.005) and Southern Africa (0.065, p = 0.042) have positive individual slopes, whereas the West, Central, and East estimates are not individually significant. Those point estimates do not establish statistically different regional mechanisms: the joint Wald test of equality across the five regional slopes has p = 0.184. We therefore treat the regional pattern as descriptive heterogeneity, not evidence of separate structural transmission regimes.

The high-risk threshold model also argues against a crisis-amplification narrative. Below the country-specific 75th percentile, the NBERI slope is 0.050 (p = 0.036). The incremental slope above the threshold is -0.030 (p = 0.339), and the total high-risk slope is not significant (p = 0.149). There is no evidence that the predictive relationship becomes systematically stronger once NBERI crosses the high-risk threshold.

4.4 Pooled panel GARCH-X: a stricter conditional-variance test

Table 5. Pooled monthly panel GARCH-X results

Model	Parameter	Estimate	SE	p-value	AIC	BIC
Panel GARCH	alpha + beta	0.995			14985.0	15012.1
Panel GARCH-X	NBERI	-0.0046	0.0106	0.663	14986.8	15020.7
Panel GARCH-X + inflation	NBERI	-0.0049	0.0107	0.647	14988.8	15029.4
Regional GARCH-X: North	NBERI	0.0120	0.0258	0.641		
Regional GARCH-X: West	NBERI	-0.0170	0.0197	0.388		
Regional GARCH-X: Central	NBERI	-0.0520	0.0301	0.084		
Regional GARCH-X: East	NBERI	0.0197	0.0225	0.381		
Regional GARCH-X: Southern	NBERI	0.0033	0.0243	0.891		

Notes: Likelihood-ratio test for GARCH versus GARCH-X: LR = 0.191, p = 0.662. Test of common versus regional NBERI variance parameters: LR = 4.644, p = 0.326. Models use within-country standardized monthly returns and Student-t innovations.

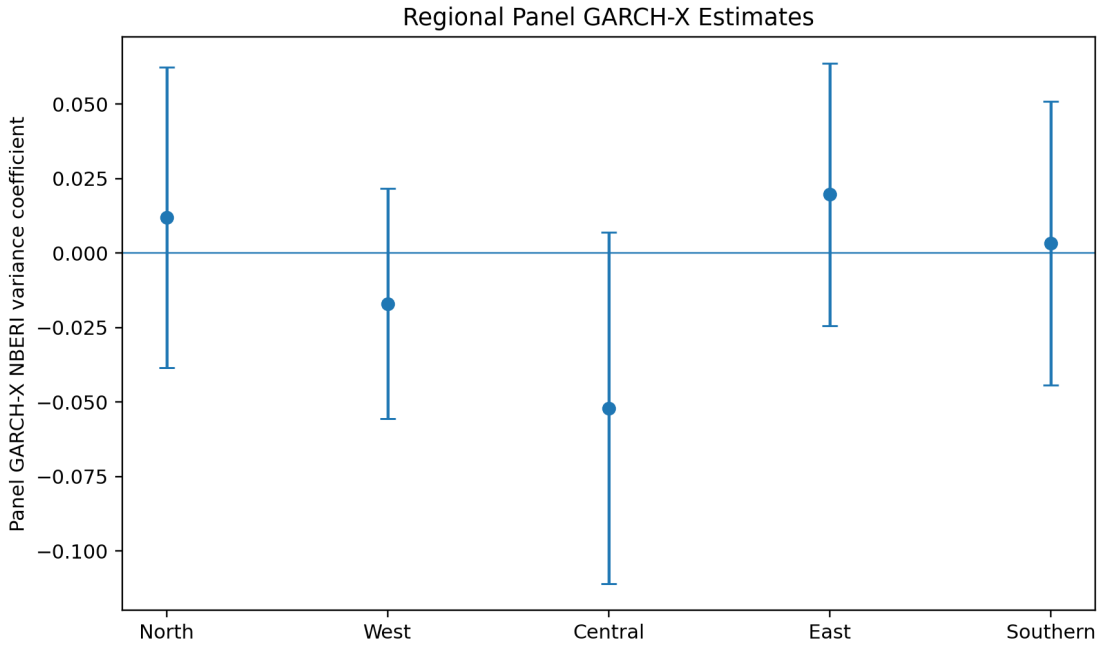


Figure 3: Regional NBERI coefficients in the pooled monthly panel GARCH-X model.

The panel GARCH exercise produces a different result from the two-way fixed-effects model. Monthly conditional variance is highly persistent: alpha + beta is approximately 0.995. Once that recursion is absorbed, the common NBERI variance coefficient is -0.0046 (p = 0.663). The likelihood-ratio test comparing GARCH with GARCH-X is insignificant (p = 0.662), and AIC and BIC do not favor the NBERI-augmented model. Adding inflation does not change the conclusion. Regional GARCH-X coefficients are individually insignificant, and the common-versus-regional likelihood-ratio test has p = 0.326.

The result narrows the mechanism. The fixed-effects model asks whether the monthly risk state contains information about the next month's observed exchange-rate movement after controls. The GARCH-X model asks whether NBERI shifts a common parametric variance recursion after lagged variance and squared innovations have absorbed nearly all persistence. The data support the first relationship, not the second. Treating those tests as interchangeable would overstate the evidence.

4.5 Dynamic feedback: PVAR, generalized responses, and variance decomposition

Table 6. Fixed-effects PVAR(1) coefficients

Equation	Lagged regressor	Coefficient	DK SE	p-value
NBERI	Lag NBERI	0.366	0.019	<0.001
NBERI	Lag FX return	0.038	0.017	0.023
NBERI	Lag FX volatility	0.036	0.017	0.031
NBERI	Lag inflation	0.020	0.013	0.109
FX return	Lag NBERI	0.025	0.010	0.016
FX return	Lag FX return	0.295	0.027	<0.001
FX return	Lag FX volatility	-0.004	0.030	0.881
FX return	Lag inflation	0.000	0.020	0.995
FX volatility	Lag NBERI	0.030	0.010	0.004
FX volatility	Lag FX return	0.075	0.032	0.022
FX volatility	Lag FX volatility	0.273	0.035	<0.001
FX volatility	Lag inflation	0.073	0.019	<0.001

Notes: All variables are standardized within country. Country and calendar-time effects are included in every equation; lagged inflation enters as a control. Maximum companion-root modulus = 0.390, satisfying the stability condition.

The PVAR reinforces the one-step panel result while clarifying feedback. Lagged NBERI predicts the FX volatility equation with a coefficient of 0.030 ($p = 0.004$), essentially matching the baseline fixed-effects estimate. NBERI also predicts next-month FX returns (0.025, $p = 0.016$). In the NBERI equation, lagged FX return and lagged FX volatility are both positive and significant at conventional levels, indicating feedback in the information environment rather than a strictly one-directional system. The NBERI autoregressive coefficient is 0.366, confirming persistence without approaching a unit root in the estimated transition matrix.

The generalized response is positive on impact and decays quickly. At horizon 1 the standardized volatility response is 0.050, with a 95 percent country-block bootstrap interval of approximately 0.026 to 0.077. At horizon 2 the response is 0.026, with an interval of roughly 0.012 to 0.043. The profile approaches zero thereafter. Because the generalized response is derived from the reduced-form covariance matrix, we read it as dynamic propagation conditional on the system, not as the effect of an externally identified structural shock.

The variance decomposition puts the size of the NBERI channel into perspective. At a 12-month horizon, NBERI innovations account for approximately 1.05 percent of the forecast-error variance of FX volatility, compared with 8.90 percent for FX-return innovations and 90.04 percent for volatility's own innovations. NBERI therefore carries statistically detectable leading information, but it is not the dominant source of exchange-rate volatility variation.

4.6 Local-projection response profile

Local projections yield the same short-horizon direction without requiring recursive PVAR propagation. The horizon-1 coefficient is 0.049 ($p < 0.001$), and the horizon-2 coefficient is 0.038 ($p < 0.001$). The response is smaller and less precise at horizons 3 and 4, becomes positive and significant again at several later horizons, and remains modest throughout. The profile is not a smooth geometric decay. The most defensible reading is an immediate and near-term predictive relationship that weakens and turns non-monotonic at longer horizons.

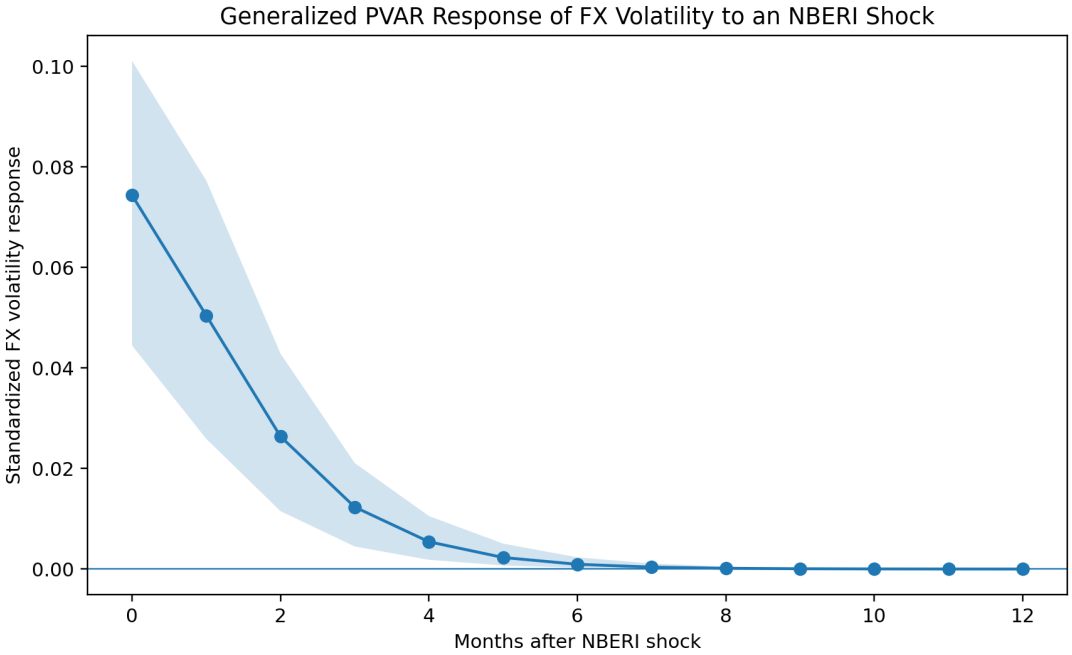


Figure 4: Generalized PVAR response of FX volatility to an NBERI innovation.

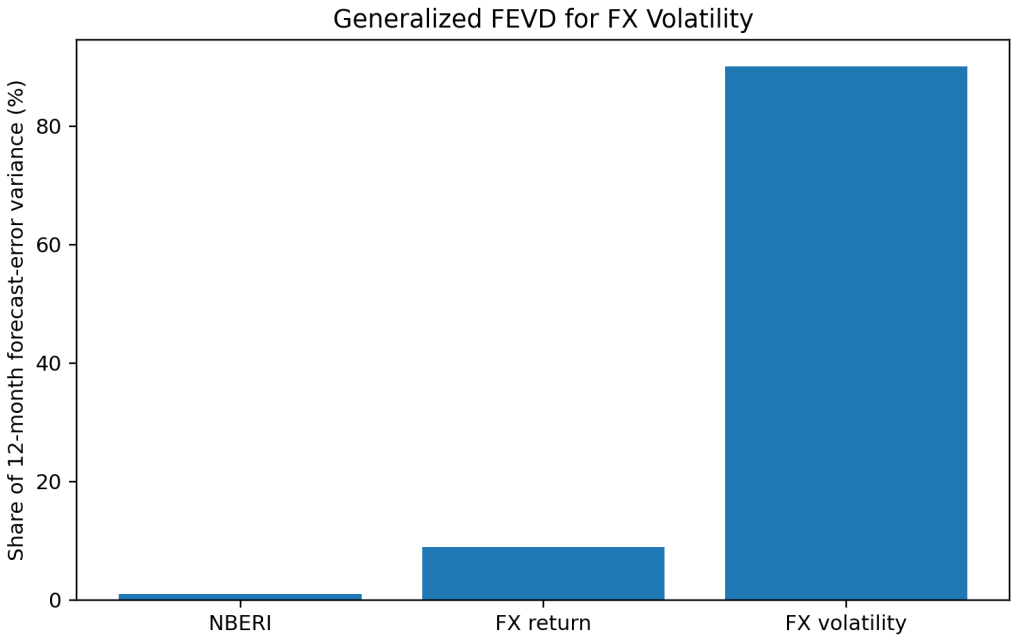


Figure 5: Generalized 12-month forecast-error variance decomposition for FX volatility.

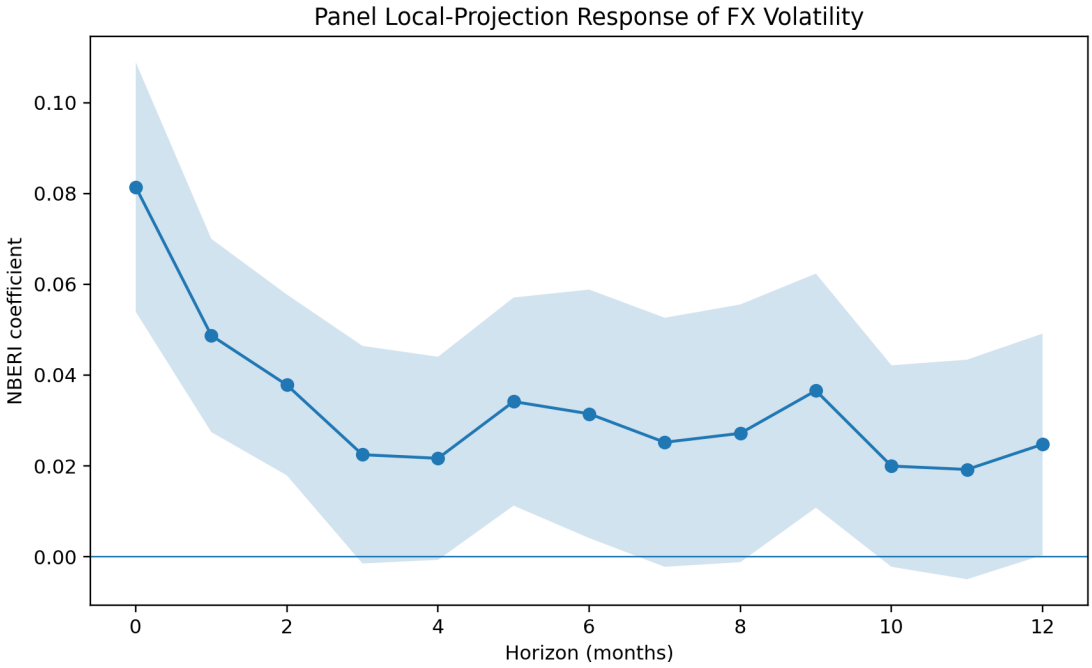


Figure 6: Panel local-projection response of the monthly FX volatility proxy to NBERI.

4.7 Out-of-sample predictive content

Table 7. Expanding-window forecast horse race, 2023-2025

Result	Statistic	RMSE / mean	QLIKE	Improvement	p-value
Pooled benchmark		2.029	32.702		
Pooled + NBERI		2.025	31.615	0.20%	
Clark-West nested test	Mean statistic	0.026			0.001
QLIKE loss-difference test	Baseline - NBERI	1.050			<0.001
Country distribution	23 of 43 positive QLIKE gains			53.5%	

Notes: The benchmark uses two lags of the log variance proxy, lagged FX return, two lags of inflation, and country intercepts. The augmented model adds two lags of NBERI. Standardization parameters are estimated using training data only at each forecast origin. Country distribution includes markets with at least 24 forecast observations.

Out-of-sample evidence supports incremental information, although the economic magnitude is modest in pooled RMSE. Adding lagged NBERI reduces pooled RMSE from 2.029 to 2.025, an improvement of approximately 0.20 percent. The Clark-West test rejects equal nested predictive accuracy in favor of the augmented model (one-sided $p = 0.0014$). QLIKE also improves in the pooled sample: the mean monthly loss difference is 1.050, with a one-sided HAC p-value below 0.001.

The gains are not uniform. Among the 43 countries with at least 24 out-of-sample observations, 23 (53.5 percent) record a positive mean QLIKE loss reduction. The cross-country distribution includes both sizable gains and deterioration. This pattern is consistent with NBERI providing useful incremental information at the continental level while its forecasting value remains market-dependent.

4.8 Robustness, persistence, and multiplicity discipline

The baseline finding is not specific to the log-squared-return transformation. Replacing the dependent variable with standardized absolute returns produces an NBERI coefficient of 0.027 ($p = 0.010$). By contrast, replacing the lagged NBERI level with its one-month first difference produces a coefficient of 0.002 ($p = 0.803$). The predictive content therefore resides primarily in the sustained risk state, not in a single-month change in the index.

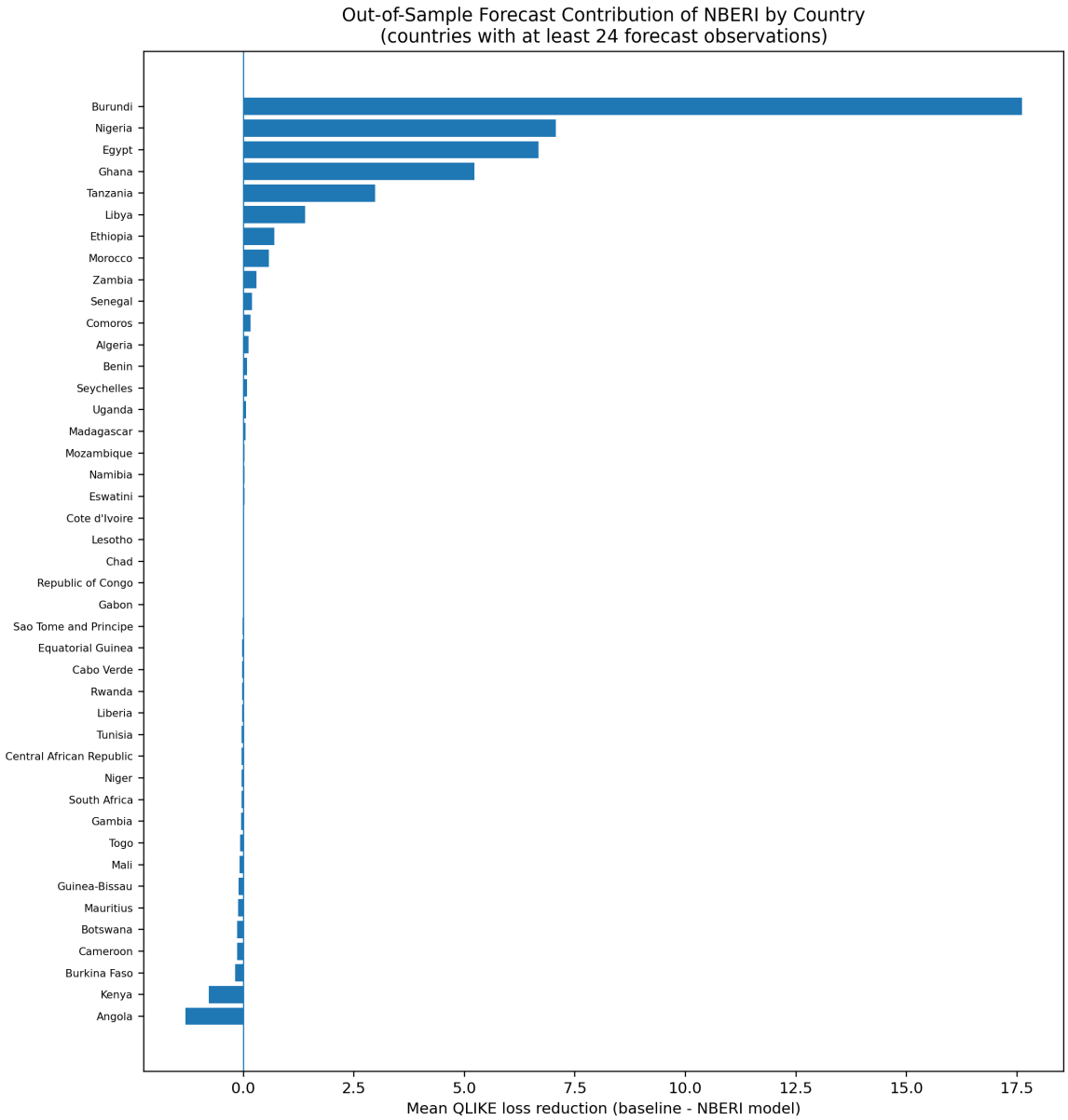


Figure 7: Country-level out-of-sample QLIKE loss reduction from adding NBERI.

4.9 FX-risk/exchange-rate alignment across the top 15 markets

Table 8. FX-risk pillar and exchange-rate-level alignment, 2015-2025

Country	Raw Pearson	Spearman	Linear-detrended log S	Cubic-detrended log S
Zambia	0.718	0.787	0.103	0.038
Ghana	0.668	0.514	0.406	0.021
Malawi	0.649	0.467	0.358	0.225
Tanzania	0.561	0.641	-0.072	0.002
Algeria	0.549	0.562	0.542	0.477
Burundi	0.480	0.376	0.241	-0.003
Libya	0.415	0.411	0.351	0.268
Nigeria	0.322	0.150	0.439	0.338
Tunisia	0.296	0.336	0.223	0.140
Morocco	0.285	0.237	0.327	0.283
Liberia	0.279	0.316	0.573	0.255
Namibia	0.268	0.287	0.434	0.443
Congo DRC	0.266	0.343	0.319	0.258
Guinea	0.263	0.382	-0.062	-0.180
Benin	0.247	0.325	0.256	0.289

Notes: Raw Pearson correlation is the ranking statistic used to define the top-15 alignment group. Detrended columns correlate residuals after removing country-specific linear or cubic time trends from log exchange rates and the FX-risk pillar. These sensitivity columns diagnose whether raw level co-movement is dominated by common trending behavior; they are not used to re-select countries.

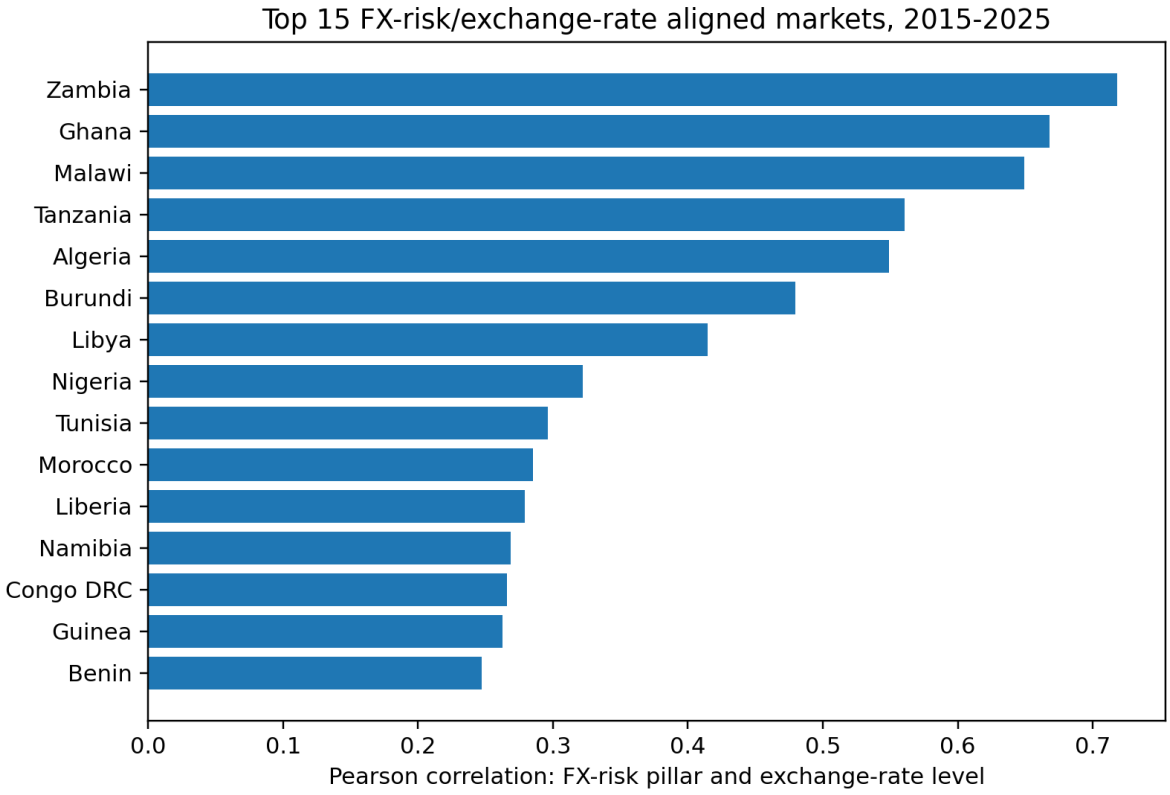


Figure 8: Top 15 FX-risk/exchange-rate aligned markets.

The alignment ranking is economically heterogeneous rather than a simple proxy for trend. Raw correlations range from 0.718 in Zambia to 0.247 in Benin. Rank correlations remain positive for all 15 markets and exceed 0.40

in several cases, while detrended correlations are strong in Algeria, Nigeria, Namibia, Liberia, Ghana, Malawi, and Libya but much weaker in Zambia, Tanzania, and Guinea. This is why the level correlation is a screening statistic rather than an identifying variable: it identifies markets in which the news-risk state and the observed FX state co-move strongly in levels, but the mechanism behind that co-movement differs across countries.

4.10 Does alignment identify stronger risk-to-volatility transmission?

Table 9. Full-sample tests of alignment heterogeneity

Test	Estimate	SE / benchmark	p-value
FE: remaining eligible slope	0.017	0.012	0.147
FE: selected-15 incremental slope	0.046	0.026	0.074
FE: selected-15 total slope	0.063	0.023	0.007
LP h=1: remaining eligible slope	0.029	0.013	0.032
LP h=1: selected-15 incremental slope	0.072	0.030	0.019
LP h=1: selected-15 total slope	0.100	0.025	0.000
Continuous NBERI x alignment interaction	0.012	0.015	0.412
Random-subset benchmark: selected total slope	0.061	94.4th percentile	0.057

Notes: Fixed-effects and local-projection interaction models are estimated in the full eligible-country panel with country and calendar-time effects and the same dynamic controls used in the benchmark. The random-subset benchmark compares the selected group with 1,000 draws of 15 eligible countries and reports an empirical one-sided probability.

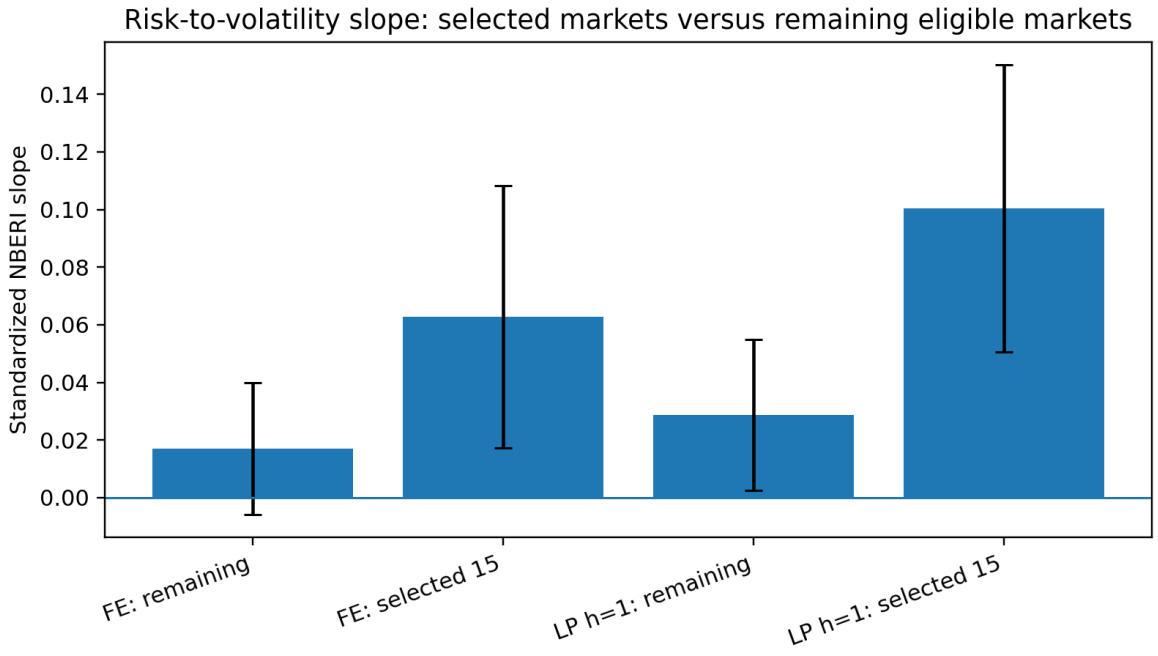


Figure 9: Risk-to-volatility slopes in aligned and remaining eligible markets.

The group comparison is the formal evidence that the alignment screen is associated with stronger transmission. In the remaining eligible markets, the lagged-NBERI coefficient is 0.017 (p = 0.147). The selected-15 total slope is 0.063 (p = 0.007), with an incremental selected-group effect of 0.046 (p = 0.074). The horizon-1 result is

sharper: the remaining-market slope is 0.029 (p = 0.032), the selected-group slope is 0.100 (p < 0.001), and the selected-group increment is 0.072 (p = 0.019). The selected-group fixed-effects slope lies near the 94th percentile of 1,000 random 15-country subsets, with a one-sided empirical probability of approximately 0.057. This is suggestive rather than decisive permutation evidence, but it shows that the top-15 result is not typical of an arbitrary 15-country draw.

The continuous interaction between NBERI and the country-level alignment coefficient is positive but insignificant (0.012, p = 0.412). Alignment therefore does not act as a linear dose-response moderator. The evidence is better described as a grouped market-state distinction: the highest-alignment markets collectively display stronger short-horizon transmission, but a larger raw level correlation does not mechanically imply a proportionally larger volatility coefficient.

4.11 Dynamic and forecast evidence in the aligned-market subsample

Table 10. Dynamic and forecast evidence in the 15 aligned markets

Result	Estimate	SE / interval	p-value
Two-way FE: lagged NBERI	0.068	0.027	0.013
PVAR volatility equation: lagged NBERI	0.068	0.027	0.013
Generalized IRF, h=1	0.099	95% CI [0.054, 0.131]	
12-month FEVD share from NBERI	3.39%		
Local projection, h=1	0.121	0.032	0.0001
Absolute-return robustness	0.066	0.029	0.023
First-difference NBERI robustness	-0.025	0.031	0.406
Forecast RMSE improvement	0.89%		
Forecast QLIKE improvement	2.85%		
Clark-West nested test	0.123		p = 0.0005
QLIKE loss-difference test	1.589		p = 0.123

The aligned-market subsample strengthens the dynamic signal without changing its interpretation. The fixed-effects coefficient is 0.068 (p = 0.013), more than twice the continental coefficient. The PVAR returns the same one-step coefficient and remains dynamically stable, with a maximum companion-root modulus of 0.483. A one-standard-deviation NBERI innovation generates a generalized volatility response of 0.099 at horizon 1, with a 95 percent country-block bootstrap interval of 0.054 to 0.131, and 0.056 at horizon 2. NBERI accounts for 3.39 percent of the 12-month volatility forecast-error variance, compared with 1.05 percent in the continental system.

Local projections show a broad near-term response: the h=1 coefficient is 0.121 (p < 0.001), h=2 is 0.130 (p < 0.001), and the response remains positive through several subsequent horizons. The absolute-return specification is also positive (0.066, p = 0.023), whereas the first difference of NBERI is insignificant (-0.025, p = 0.406). As in the continental results, the information appears to reside in a persistent risk state rather than a one-month jump in the index.

4.12 Country-level GARCH-family evidence and out-of-sample gains

Table 11. Profile-likelihood GARCH-family results surviving within-family FDR correction

Country	Model	delta	LR p	BH q	LB(5) sq. p	ARCH-LM(5) p
Ghana	EGARCH-X	0.766	0.0002	0.0036	0.044	0.078
Libya	GARCH-X	0.645	0.0005	0.0074	1.000	1.000

Notes: Reported rows are the only country-family X effects that survive Benjamini-Hochberg q < 0.10 after profile-likelihood refinement of the NBERI variance coefficient. Forty-five X models are tested in total: GARCH-X, EGARCH-X, and GJR/TGARCH-X for each of 15 countries. Full results are reported in Appendix H.

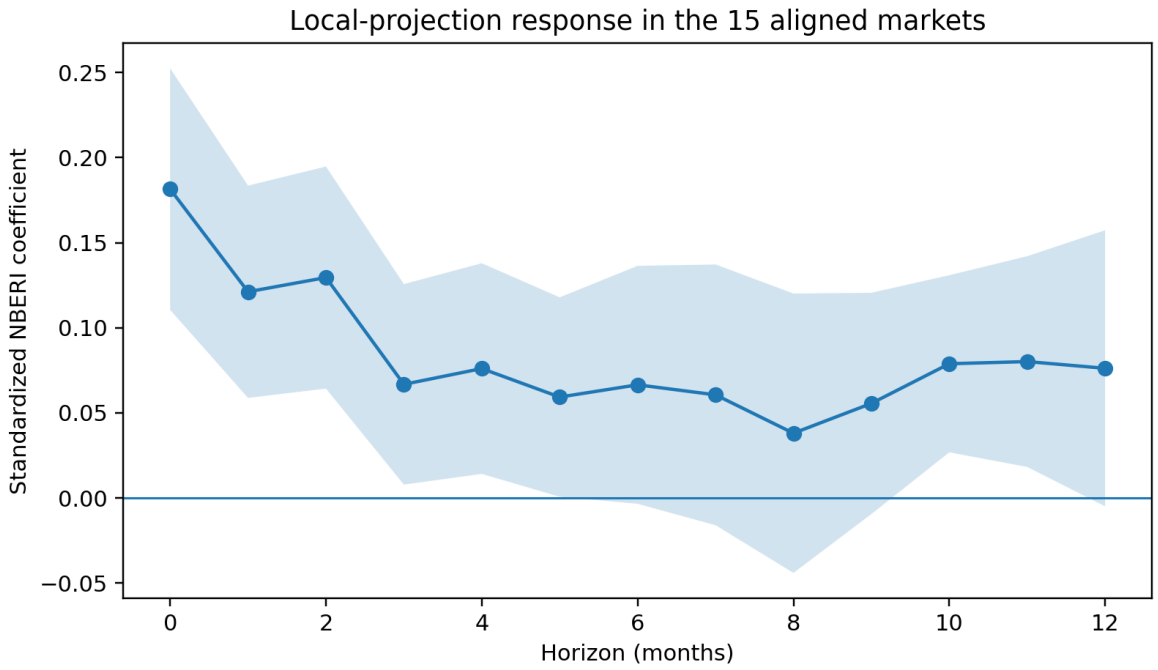


Figure 10: Local-projection response in the 15 aligned markets.

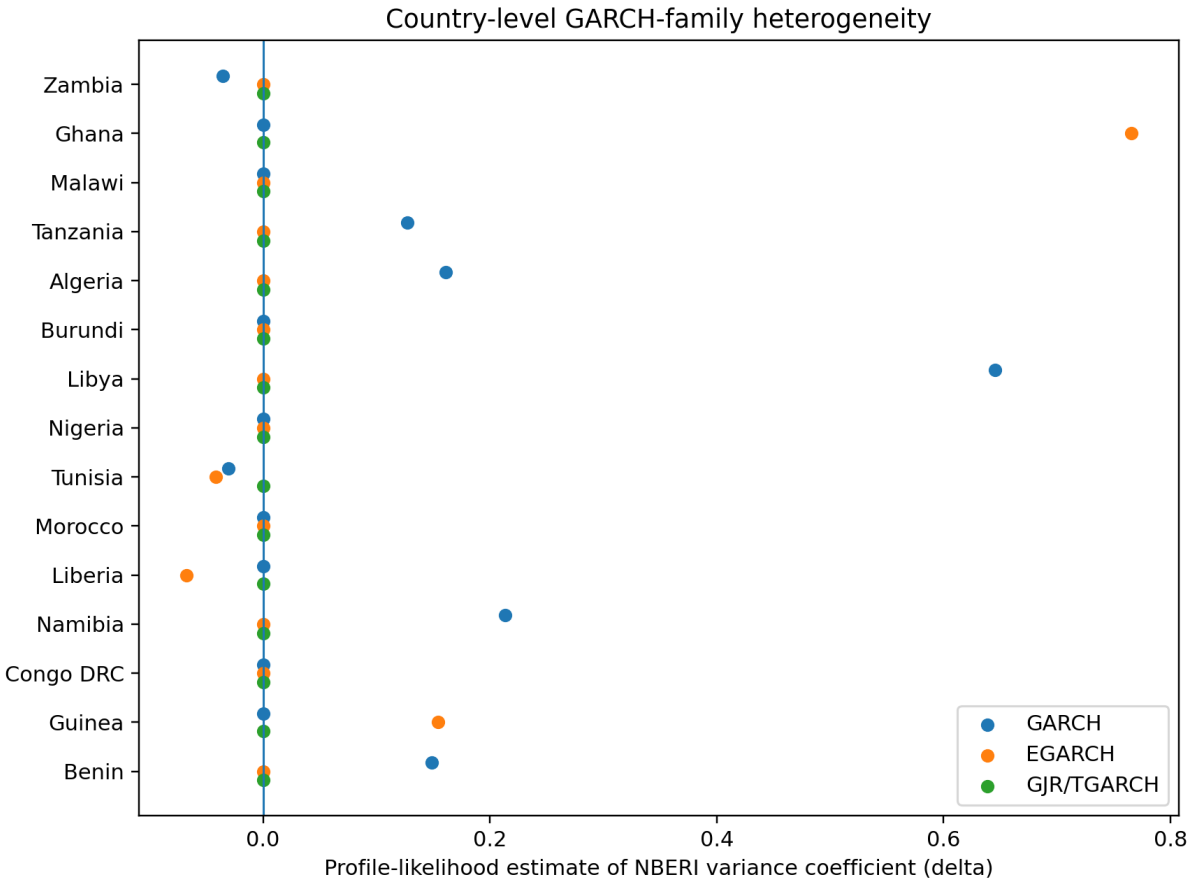


Figure 11: Profile-likelihood NBERI variance coefficients across the 15-country GARCH family.

The profile-likelihood exercise sharply narrows the country-level GARCH evidence. Only two effects survive within-family false-discovery control: Ghana under EGARCH-X (delta = 0.766; LR p = 0.00024; q = 0.0036) and Libya under GARCH-X (delta = 0.645; LR p = 0.00050; q = 0.0074). Libya passes the squared-residual and ARCH diagnostics comfortably. Ghana's ARCH-LM diagnostic is acceptable at conventional levels, although the Ljung-Box test on squared standardized residuals is borderline at p = 0.044; we therefore interpret the Ghana result with additional caution. The remaining 43 country-family X tests do not survive q < 0.10 after the profile-likelihood refinement.[11] The pattern is more informative than a list of nominal country p-values: it shows that conditional-variance transmission exists in specific markets but is not a general feature of the aligned sample.

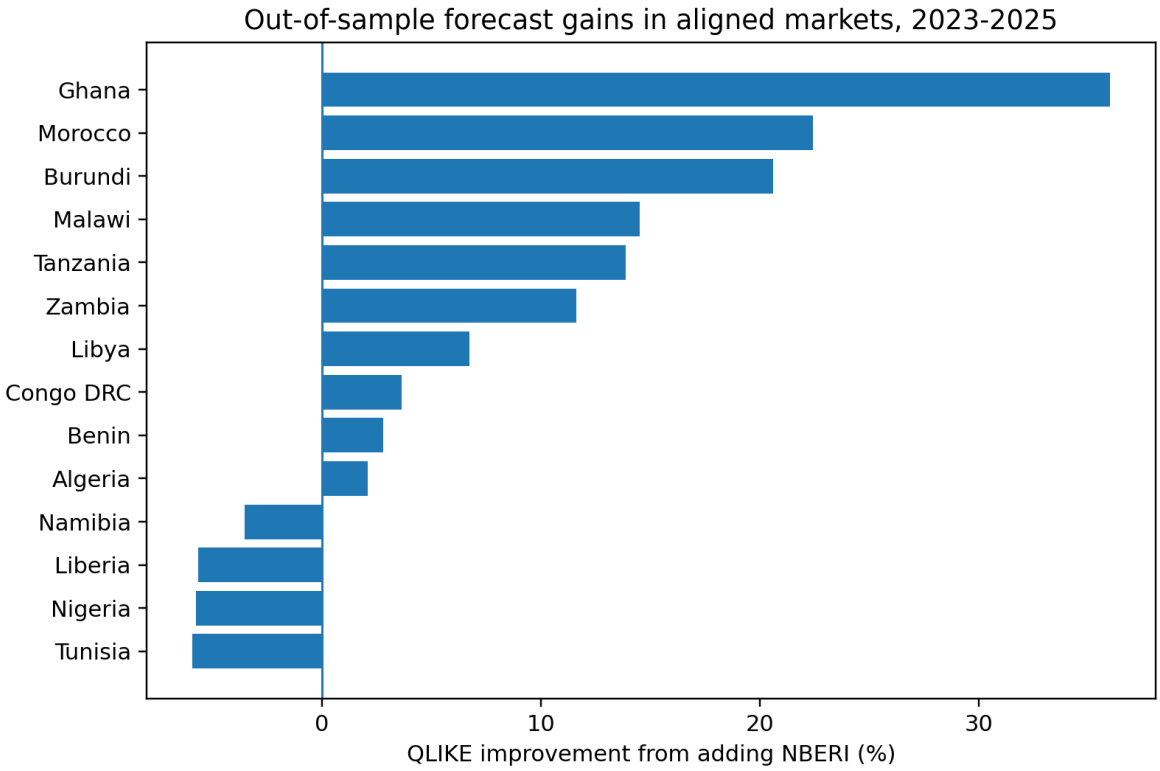


Figure 12: Country-level QLIKE improvement from adding NBERI in the aligned-market forecast experiment.

The forecast horse race adds a genuine prediction test. Across the aligned markets, adding lagged NBERI lowers pooled RMSE by 0.89 percent, from 2.254 to 2.234, and the Clark-West test favors the augmented model (p = 0.0005). Mean QLIKE falls by 2.85 percent, although the HAC test of the pooled QLIKE loss difference is not significant at conventional levels (p = 0.123). The country distribution is heterogeneous: Ghana records the largest QLIKE improvement, followed by Morocco and Burundi, while several countries show small deteriorations. The forecast evidence therefore supports incremental predictive content without implying uniform country-level dominance.

5. Discussion, Conclusion, and Limitations

5.1 Continental leading information and the boundary of the claim

The continental evidence establishes the baseline result: NBERI contains leading information about subsequent monthly exchange-rate instability. The coefficient survives market-history controls, inflation, country effects, common time shocks, and the collapse of repeated currency-union paths. PVARs, generalized responses, local

[11]The false-discovery-rate adjustment is applied within model family across the 15 aligned countries. This is stricter than highlighting isolated nominal p-values from dozens of country-model combinations.

projections, absolute-return robustness, and out-of-sample tests all point the same way. The size is economically modest, as expected for an incremental information variable rather than a stand-alone market model.

The pooled GARCH-X null is a limit worth stating clearly. The common monthly variance recursion is very persistent and does not admit a statistically meaningful continent-wide NBERI variance coefficient. The country-family exercise does not overturn that result; it explains it. After profile-likelihood refinement and multiplicity control, only Ghana and Libya exhibit robust family-specific X effects. The absence of a universal variance parameter is therefore not evidence that NBERI lacks information. It indicates that the information enters market dynamics through heterogeneous channels rather than one common GARCH mechanism.

5.2 Market alignment as an organizing device for heterogeneity

The market-alignment extension is the paper's main heterogeneity result. The top-15 aligned markets have a larger total fixed-effects slope than the remaining eligible countries and a significantly larger horizon-1 local-projection response. The selected-group coefficient also lies near the 94th percentile of random 15-country subsets. At the same time, the continuous alignment interaction is insignificant, so the raw correlation should not be interpreted as a structural moderator measured on a cardinal scale. The useful insight is categorical and empirical: markets in which FX-risk narratives and the observed exchange-rate state are most tightly synchronized, as a group, transmit the news-risk state into subsequent volatility more strongly.

5.3 Why the GARCH-family and forecast results matter

The combined evidence separates three objects that are often conflated: information, conditional variance, and forecasting value. NBERI is informative in the panel and local-projection sense; conditional-variance effects are concentrated in a small number of countries; and out-of-sample gains are positive on average but uneven across markets and loss functions. This distinction is operationally useful. A surveillance system can treat NBERI as a state variable that enriches the information set without assuming that it enters every currency's variance equation or improves every forecast origin.

5.4 Contribution to empirical FX-risk measurement

The paper's empirical design for evaluating text-based market-risk measures proceeds in stages: build a broad panel, keep news risk separate from market outcomes, test effective currency-market dependence, distinguish predictive regressions from conditional-variance recursions, trace dynamic propagation, run a real-time forecast evaluation, and use an alignment screen only as a heterogeneity extension. The resulting conclusion is stronger because it is narrower: NBERI is a leading information state with economically meaningful market heterogeneity, not a universal structural volatility factor.

5.5 Conclusion

This paper evaluates a news-based exchange-rate risk measure across the broadest African sample supported by the supplied monthly data and adds a new market-alignment framework for understanding why the signal is stronger in some currency markets than others. Of 54 source countries, 51 pass pre-specified coverage and variation screens. The design keeps the news measure separate from market outcomes, adjusts inference for serial and cross-sectional dependence, collapses shared-currency paths, and combines panel prediction, dynamic propagation, conditional-variance models, and real-time forecast evaluation in one framework.

The continental result is a positive but moderate leading relationship: a one-standard-deviation increase in lagged NBERI predicts approximately 0.03 standard deviations more next-month FX volatility after controls, and the relationship survives the 36-cluster currency-market robustness. The alignment extension adds a second result. In the top 15 FX-risk/exchange-rate aligned markets, the fixed-effects coefficient is 0.068 ($p = 0.013$), the horizon-1 local-projection response is 0.121 ($p < 0.001$), and NBERI explains 3.39 percent of the 12-month volatility forecast-error variance. Full-sample interaction tests show that the aligned group has a significantly larger horizon-1 slope than the remaining eligible markets.

Out-of-sample performance is consistent with incremental rather than dominant predictive value. In the aligned markets, NBERI lowers pooled RMSE by 0.89 percent and passes the Clark-West nested forecast test, while the pooled QLIKE improvement is positive but not statistically decisive under the HAC loss-difference test. The country-level GARCH family provides an equally important limit: after profile-likelihood refinement and false-discovery correction, robust NBERI variance effects survive only in Ghana and Libya. The common pooled GARCH-X null and the selective country-family evidence are therefore mutually consistent.

The novelty is the separation of broad informational content from market-specific volatility transmission. News-based FX risk contains measurable leading information across the continent, but the strength and mechanism of that information depend on market context. The market-alignment layer identifies a group with stronger short-horizon transmission without converting a descriptive level correlation into a causal claim. For surveillance, this implies that text-based FX-risk measures are most useful as disciplined additions to the market information set, with country-specific model validation determining when they should enter conditional-variance or forecasting systems.

5.6 Limitations

The main limitation is data frequency. Monthly exchange rates cannot recover intramonth quadratic variation, so the panel outcome is a monthly volatility proxy and the GARCH-family models have relatively short time series. This is why the panel, dynamic, and forecast evidence carries more weight than any single country-level variance recursion. Higher-frequency exchange-rate data would permit sharper GARCH-X, EGARCH-X, and threshold-volatility tests.

A second limitation concerns the market-alignment screen. Exchange-rate levels can be persistent or trending, and a contemporaneous level correlation is not an identification strategy. We treat the top-15 ranking as descriptive, report rank and detrended sensitivity measures, run full-sample interaction tests, and compare the selected group with random 15-country subsets. These checks strengthen the heterogeneity interpretation but do not make the alignment coefficient causal.

Finally, NBERI is observational and the 2023-2025 forecast evaluation window is short relative to the full estimation period. Lagging the index, fixed effects, reverse PVAR paths, state-versus-change robustness, profile-likelihood volatility tests, multiplicity control, and genuine out-of-sample evaluation narrow several alternative explanations but do not identify an exogenous structural shock. We therefore maintain predictive language and treat country-specific mechanisms as hypotheses for future work with higher-frequency prices and richer institutional covariates.

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A. Country Coverage and Screening

The table reports the pre-model coverage screen for every country in the source exchange-rate panel. Eligibility requires at least 60 usable adjacent monthly returns, at least 5 percent nonzero monthly changes, and nonzero return variance.

Table A1. Country coverage screen

Country	Region	Usable returns	Nonzero	Return SD	Eligible	Currency cluster
Algeria	North	131	100.0%	0.973	Yes	Algeria
Angola	Central	131	97.7%	4.495	Yes	Angola
Benin	West	131	100.0%	1.673	Yes	WAEMU-XOF
Botswana	Southern	131	100.0%	2.886	Yes	Botswana
Burkina Faso	West	131	100.0%	1.673	Yes	WAEMU-XOF
Burundi	East	125	100.0%	2.368	Yes	Burundi
Cabo Verde	West	131	99.2%	1.716	Yes	Cabo Verde
Cameroon	Central	131	100.0%	1.673	Yes	CEMAC-XAF
Central African Republic	Central	131	100.0%	1.673	Yes	CEMAC-XAF
Chad	Central	131	100.0%	1.673	Yes	CEMAC-XAF
Comoros	East	131	100.0%	1.673	Yes	Comoros
DR Congo	Central	127	100.0%	2.513	Yes	DR Congo

Country	Region	Usable returns	Nonzero	Return SD	Eligible	Currency cluster
Republic of Congo	Central	131	100.0%	1.673	Yes	CEMAC-XAF
Cote d'Ivoire	West	131	100.0%	1.673	Yes	WAEMU-XOF
Djibouti	East	131	0.0%	0.000	No	Djibouti
Egypt	North	131	99.2%	6.715	Yes	Egypt
Equatorial Guinea	Central	131	100.0%	1.673	Yes	CEMAC-XAF
Eritrea	East	131	0.8%	0.172	No	Eritrea
Eswatini	Southern	131	100.0%	3.339	Yes	CMA-ZAR-linked
Ethiopia	East	128	100.0%	5.150	Yes	Ethiopia
Gabon	Central	131	100.0%	1.673	Yes	CEMAC-XAF
Gambia	West	131	100.0%	1.976	Yes	Gambia
Ghana	West	131	100.0%	5.331	Yes	Ghana
Guinea	West	76	100.0%	1.117	Yes	Guinea
Guinea-Bissau	West	131	100.0%	1.673	Yes	WAEMU-XOF
Kenya	East	131	100.0%	1.498	Yes	Kenya
Lesotho	Southern	131	100.0%	3.339	Yes	CMA-ZAR-linked
Liberia	West	127	96.1%	2.841	Yes	Liberia
Libya	North	130	100.0%	10.560	Yes	Libya
Madagascar	East	131	100.0%	1.836	Yes	Madagascar
Malawi	Southern	116	100.0%	3.835	Yes	Malawi
Mali	West	131	100.0%	1.673	Yes	WAEMU-XOF
Mauritania	West	115	97.4%	0.985	Yes	Mauritania
Mauritius	East	131	100.0%	1.322	Yes	Mauritius
Morocco	North	131	100.0%	1.377	Yes	Morocco
Mozambique	Southern	127	74.0%	3.271	Yes	Mozambique
Namibia	Southern	131	100.0%	3.354	Yes	CMA-ZAR-linked
Niger	West	131	100.0%	1.673	Yes	WAEMU-XOF
Nigeria	West	131	87.0%	6.399	Yes	Nigeria
Rwanda	East	131	100.0%	0.412	Yes	Rwanda
Sao Tome and Principe	Central	128	99.2%	1.687	Yes	Sao Tome and Principe
Senegal	West	131	100.0%	1.673	Yes	WAEMU-XOF
Seychelles	East	131	100.0%	2.977	Yes	Seychelles
Sierra Leone	West	108	100.0%	2.190	Yes	Sierra Leone
Somalia	East	41	97.6%	2.414	No	Somalia
South Africa	Southern	131	100.0%	3.339	Yes	CMA-ZAR-linked
South Sudan	East	124	91.9%	14.231	Yes	South Sudan
Sudan	North	97	42.3%	17.465	Yes	Sudan
Tanzania	East	123	99.2%	1.311	Yes	Tanzania
Togo	West	131	100.0%	1.673	Yes	WAEMU-XOF
Tunisia	North	129	100.0%	1.488	Yes	Tunisia
Uganda	East	131	100.0%	1.510	Yes	Uganda
Zambia	Southern	131	99.2%	5.235	Yes	Zambia
Zimbabwe	Southern	81	98.8%	20.503	Yes	Zimbabwe

B. Diagnostics

Table B1. Panel diagnostic summary

Variable	Countries	ADF rejects	ADF %	KPSS rejects	KPSS %
NBERI	51	33	64.7%	25	49.0%
FX return	51	46	90.2%	1	2.0%
Volatility proxy	51	46	90.2%	9	17.6%

Variable	Countries	ADF rejects	ADF %	KPSS rejects	KPSS %
ARCH-LM(5): FX return	51	13	25.5%		

Notes: ADF null: unit root. KPSS null: level stationarity. ARCH-LM row reports rejection of no ARCH at 5 percent under the ADF columns for compactness. Tests are country-specific.

C. Full Local-Projection Horizon Estimates

Table C1. Local-projection estimates

Horizon	NBERI beta	DK SE	p-value	95% low	95% high	N
0	0.081	0.014	<0.001	0.054	0.109	6,340
1	0.049	0.011	<0.001	0.027	0.070	6,297
2	0.038	0.010	<0.001	0.018	0.058	6,249
3	0.022	0.012	0.066	-0.001	0.046	6,200
4	0.022	0.011	0.057	-0.001	0.044	6,151
5	0.034	0.012	0.003	0.011	0.057	6,102
6	0.031	0.014	0.024	0.004	0.059	6,053
7	0.025	0.014	0.071	-0.002	0.053	6,004
8	0.027	0.014	0.060	-0.001	0.056	5,956
9	0.037	0.013	0.005	0.011	0.062	5,907
10	0.020	0.011	0.077	-0.002	0.042	5,859
11	0.019	0.012	0.119	-0.005	0.043	5,810
12	0.025	0.012	0.046	0.000	0.049	5,761

D. Country-Specific Dynamic Slopes

Country-specific coefficients are diagnostic rather than the basis for the paper's main inference. Benjamini-Hochberg q-values control the false discovery rate across 51 country tests.

Table D1. Country-specific dynamic NBERI slopes

Country	Region	N	Beta	HAC SE	p-value	BH q
Angola	Central	130	-0.017	0.062	0.786	0.871
Cameroon	Central	130	0.043	0.091	0.633	0.832
Central African Republic	Central	126	-0.104	0.039	0.007	0.152
Chad	Central	130	0.006	0.046	0.905	0.937
DR Congo	Central	114	-0.126	0.055	0.022	0.183
Equatorial Guinea	Central	130	0.095	0.048	0.045	0.253
Gabon	Central	130	0.104	0.067	0.122	0.430
Republic of Congo	Central	126	0.062	0.081	0.445	0.732
Sao Tome and Principe	Central	127	-0.112	0.050	0.027	0.194
Burundi	East	124	0.042	0.041	0.311	0.621
Comoros	East	129	0.007	0.074	0.924	0.937
Ethiopia	East	127	0.051	0.055	0.353	0.621
Kenya	East	130	0.055	0.049	0.261	0.609
Madagascar	East	129	0.097	0.104	0.349	0.621
Mauritius	East	130	-0.116	0.061	0.058	0.268
Rwanda	East	130	-0.087	0.037	0.018	0.181
Seychelles	East	130	0.036	0.031	0.256	0.609
South Sudan	East	114	0.053	0.091	0.564	0.822
Tanzania	East	122	0.037	0.090	0.683	0.832
Uganda	East	130	-0.068	0.071	0.342	0.621

Country	Region	N	Beta	HAC SE	p-value	BH q
Algeria	North	130	0.015	0.079	0.848	0.921
Egypt	North	130	0.275	0.151	0.069	0.293
Libya	North	125	0.155	0.059	0.009	0.152
Morocco	North	130	0.158	0.076	0.039	0.247
Sudan	North	96	0.131	0.077	0.089	0.351
Tunisia	North	128	0.011	0.090	0.905	0.937
Botswana	Southern	130	-0.024	0.072	0.737	0.835
Eswatini	Southern	130	-0.049	0.104	0.640	0.832
Lesotho	Southern	130	0.174	0.092	0.057	0.268
Malawi	Southern	115	0.082	0.183	0.654	0.832
Mozambique	Southern	126	0.063	0.095	0.505	0.773
Namibia	Southern	130	0.086	0.089	0.334	0.621
South Africa	Southern	130	0.095	0.088	0.276	0.613
Zambia	Southern	130	-0.033	0.091	0.718	0.832
Zimbabwe	Southern	68	0.129	0.154	0.402	0.683
Benin	West	130	0.066	0.101	0.515	0.773
Burkina Faso	West	130	0.008	0.095	0.937	0.937
Cabo Verde	West	130	-0.075	0.071	0.292	0.620
Cote d'Ivoire	West	130	0.124	0.082	0.131	0.430
Gambia	West	130	0.062	0.157	0.694	0.832
Ghana	West	130	0.275	0.111	0.013	0.166
Guinea	West	75	-0.123	0.107	0.249	0.609
Guinea-Bissau	West	130	0.148	0.112	0.188	0.531
Liberia	West	126	-0.145	0.130	0.263	0.609
Mali	West	129	-0.061	0.087	0.480	0.764
Mauritania	West	114	-0.020	0.013	0.135	0.430
Niger	West	130	-0.192	0.056	<0.001	0.032
Nigeria	West	130	0.066	0.122	0.587	0.832
Senegal	West	130	-0.047	0.092	0.606	0.832
Sierra Leone	West	107	-0.100	0.072	0.165	0.494
Togo	West	130	0.035	0.091	0.702	0.832

E. Conditional-Variance Robustness

A pooled GJR-GARCH-X model tests whether a common sign-dependent innovation response alters the NBERI conclusion. The asymmetry parameter is effectively zero, and the NBERI coefficient remains statistically insignificant. This result does not support a common leverage channel in the monthly panel.

Table E1. Pooled GJR-GARCH-X robustness

Model	Parameter	Estimate	SE	p-value	AIC	BIC
Panel GJR-GARCH	omega	0.0289			14986.9873	15020.8434
Panel GJR-GARCH	alpha	0.3530			14986.9873	15020.8434
Panel GJR-GARCH	gamma_asymmetry	0.0000			14986.9873	15020.8434
Panel GJR-GARCH	beta	0.6420			14986.9873	15020.8434
Panel GJR-GARCH	Student_t_df	4.1409			14986.9873	15020.8434
Panel GJR-GARCH-X	omega	0.0288			14988.7966	15029.4239
Panel GJR-GARCH-X	alpha	0.3533			14988.7966	15029.4239
Panel GJR-GARCH-X	gamma_asymmetry	0.0000			14988.7966	15029.4239

Model	Parameter	Estimate	SE	p-value	AIC	BIC
Panel GJR-GARCH-X	beta	0.6417			14988.7966	15029.4239
Panel GJR-GARCH-X	delta_NBERI	-0.0046	0.011	0.663	14988.7966	15029.4239
Panel GJR-GARCH-X	Student_t_df	4.1419			14988.7966	15029.4239

F. Reproducibility Definitions

The principal analysis uses the following fixed definitions: (i) monthly FX return equals 100 times the first log difference of local currency per U.S. dollar; (ii) the monthly volatility proxy equals $\ln(1 + r^2)$; (iii) pooled variables are standardized within country; (iv) the principal dynamic panel uses one lag of NBERI, volatility, return, and inflation; (v) Driscoll-Kraay bandwidth is four months; (vi) the shared-currency robustness reduces 51 country series to 36 market clusters; and (vii) the forecast horse race uses an expanding window with training-only standardization at every forecast origin. These definitions are fixed across the reported tables and figures.

G. Market-Alignment Robustness

The market-alignment ranking is descriptive and is not used as a causal identifying assumption. Table G1 reports rank-based and detrended sensitivity measures for the selected countries. Positive raw correlation is common across the group, but detrended associations vary, which is consistent with using alignment as a heterogeneity screen rather than a structural parameter.

Table G1. Market-alignment sensitivity diagnostics

Country	Raw Pearson	Spearman	Linear detrended	Cubic detrended
Zambia	0.718	0.787	0.103	0.038
Ghana	0.668	0.514	0.406	0.021
Malawi	0.649	0.467	0.358	0.225
Tanzania	0.561	0.641	-0.072	0.002
Algeria	0.549	0.562	0.542	0.477
Burundi	0.480	0.376	0.241	-0.003
Libya	0.415	0.411	0.351	0.268
Nigeria	0.322	0.150	0.439	0.338
Tunisia	0.296	0.336	0.223	0.140
Morocco	0.285	0.237	0.327	0.283
Liberia	0.279	0.316	0.573	0.255
Namibia	0.268	0.287	0.434	0.443
Congo DRC	0.266	0.343	0.319	0.258
Guinea	0.263	0.382	-0.062	-0.180
Benin	0.247	0.325	0.256	0.289

H. Full Profile-Likelihood GARCH-Family Results

Tables H1a-H1c report all 45 country-family X tests after profile-likelihood refinement. BH q-values are computed separately within GARCH-X, EGARCH-X, and GJR/TGARCH-X across the 15 countries.

Table H1a. GARCH-X country results

Country	delta	LR p	BH q	AIC	LB sq. p	ARCH-LM p	Profile grid
Algeria	0.161	0.4252	1.0000	311.1	0.976	0.980	0.15
Benin	0.149	0.4531	1.0000	502.2	0.240	0.272	0.15
Burundi	0.000	1.0000	1.0000	-346.0	1.000	1.000	-0.50
Congo DRC	0.000	1.0000	1.0000	285.4	0.967	0.975	0.75
Ghana	0.000	1.0000	1.0000	624.9	0.017	0.024	1.00

Country	delta	LR p	BH q	AIC	LB sq. p	ARCH-LM p	Profile grid
Guinea	0.000	1.0000	1.0000	142.5	0.471	0.422	0.00
Liberia	0.000	1.0000	1.0000	500.0	0.886	0.899	0.50
Libya	0.645	0.0005	0.0074	375.9	1.000	1.000	0.75
Malawi	0.000	1.0000	1.0000	278.1	1.000	1.000	0.50
Morocco	0.000	1.0000	1.0000	433.3	0.786	0.732	0.15
Namibia	0.214	0.1639	1.0000	686.2	0.983	0.986	0.15
Nigeria	0.000	1.0000	1.0000	440.3	0.997	0.997	0.30
Tanzania	0.127	0.2634	1.0000	-4.6	0.998	0.999	0.15
Tunisia	-0.031	0.7384	1.0000	453.3	0.841	0.856	0.00
Zambia	-0.036	0.7422	1.0000	748.8	0.954	0.945	0.00

Table H1b. EGARCH-X country results

Country	delta	LR p	BH q	AIC	LB sq. p	ARCH-LM p	Profile grid
Algeria	0.000	1.0000	1.0000	325.3	0.968	0.970	0.00
Benin	0.000	1.0000	1.0000	502.3	0.382	0.337	0.00
Burundi	0.000	1.0000	1.0000	-138.5	1.000	1.000	1.00
Congo DRC	0.000	1.0000	1.0000	223.5	0.983	0.979	-0.30
Ghana	0.766	0.0002	0.0036	574.0	0.044	0.078	1.00
Guinea	0.154	0.5751	1.0000	124.5	0.856	0.737	0.15
Liberia	-0.068	0.4264	1.0000	466.7	0.943	0.930	0.00
Libya	0.000	1.0000	1.0000	467.6	1.000	1.000	0.75
Malawi	0.000	1.0000	1.0000	519.0	0.999	0.999	1.00
Morocco	0.000	1.0000	1.0000	436.1	0.301	0.167	0.15
Namibia	0.000	1.0000	1.0000	685.7	0.454	0.568	0.00
Nigeria	0.000	1.0000	1.0000	731.2	0.998	0.998	1.00
Tanzania	0.000	1.0000	1.0000	218.9	0.458	0.013	0.30
Tunisia	-0.042	0.3696	1.0000	454.2	0.765	0.793	0.00
Zambia	0.000	1.0000	1.0000	767.6	0.992	0.993	0.00

Table H1c. GJR/TGARCH-X country results

Country	delta	LR p	BH q	AIC	LB sq. p	ARCH-LM p	Profile grid
Algeria	0.000	1.0000	1.0000	337.8	0.989	0.990	0.00
Benin	0.000	1.0000	1.0000	505.7	0.127	0.155	0.00
Burundi	0.000	1.0000	1.0000	335.9	1.000	1.000	0.00
Congo DRC	0.000	1.0000	1.0000	429.4	0.956	0.957	0.00
Ghana	0.000	1.0000	1.0000	695.2	0.775	0.721	0.75
Guinea	0.000	1.0000	1.0000	143.7	0.484	0.417	0.30
Liberia	0.000	1.0000	1.0000	502.4	0.868	0.874	0.00
Libya	0.000	1.0000	1.0000	745.5	1.000	1.000	0.00
Malawi	0.000	1.0000	1.0000	516.8	0.999	0.999	-0.15
Morocco	0.000	1.0000	1.0000	440.3	0.180	0.061	0.30
Namibia	0.000	1.0000	1.0000	689.7	0.543	0.640	0.00
Nigeria	0.000	1.0000	1.0000	726.4	0.998	0.998	1.00
Tanzania	0.000	1.0000	1.0000	51.4	0.977	0.880	0.15
Tunisia	0.000	1.0000	1.0000	460.8	0.802	0.836	0.00
Zambia	0.000	1.0000	1.0000	771.1	0.966	0.958	0.30

I. Random-Subset Benchmark

To assess whether the selected-15 fixed-effects slope is unusually large relative to arbitrary country groupings, 1,000 subsets of 15 countries are drawn without replacement from the 51 eligible markets. The selected group's incremental slope is at the 94.1st percentile of the random distribution and its total slope is at the 94.4th percentile.

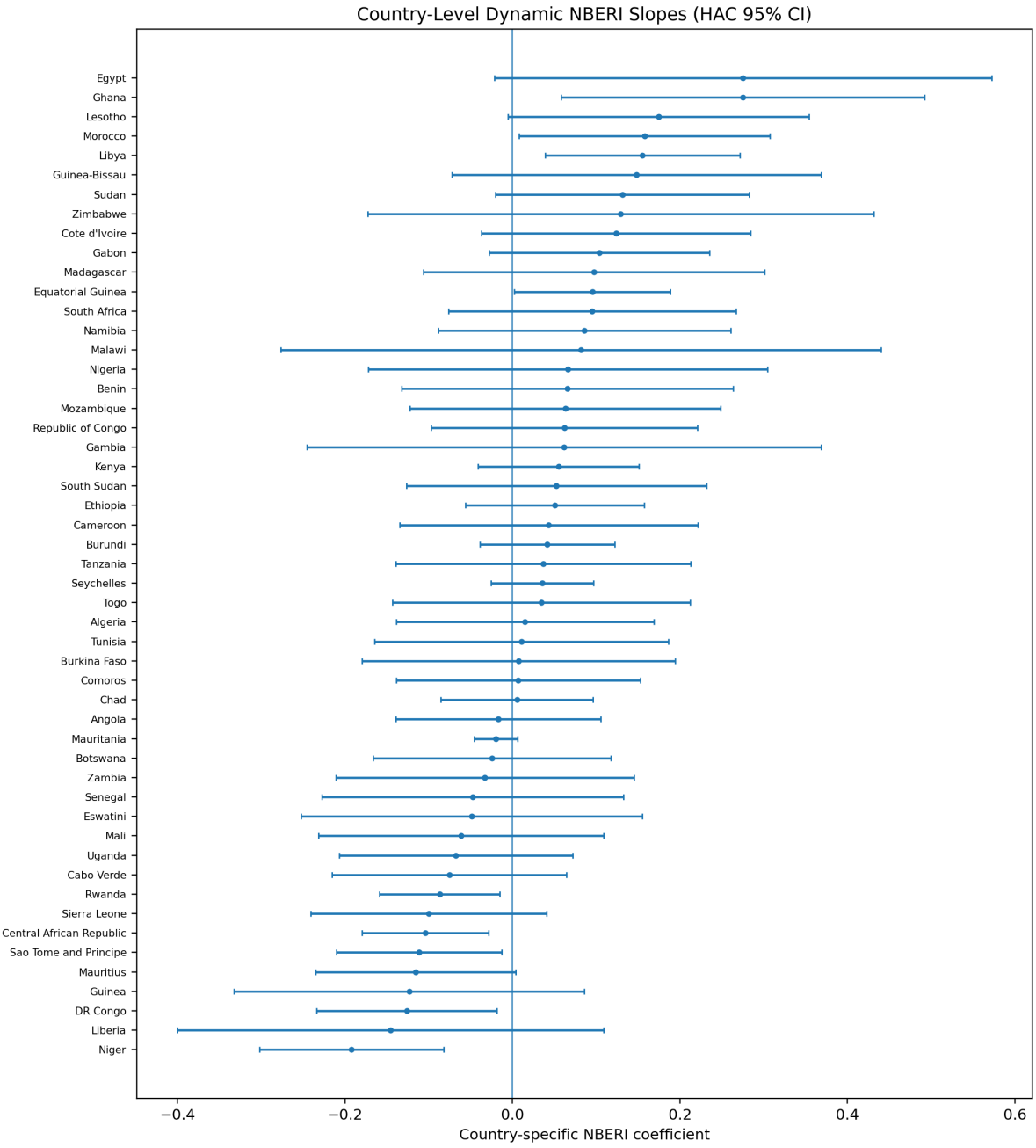


Figure 13: Country-level dynamic NBERI slopes with HAC 95 percent confidence intervals.

The corresponding one-sided empirical probabilities are 0.060 and 0.057. These values are suggestive rather than conventionally significant and are interpreted accordingly.

Table I1. Random 15-country subset benchmark

Statistic	Observed	Random mean	Random SD	Percentile	Empirical p
Selected-15 incremental slope	0.044	-0.001	0.030	94.1	0.060
Selected-15 total slope	0.061	0.028	0.021	94.4	0.057